```
clear; clc; close all;
freqGHz = [0.7 3.5 28];
freqMHz = freqGHz * 1000;
bandName = ["Sub-1 GHz"; "Mid-band"; "mmWave"];
txPowerdBm = 30;
gtDbi = 0; grDbi = 0;
distanceKm = logspace(log10(0.1),log10(2),200);
distanceAtPointKm = 0.5;
fspl = @(fMHz,dKm) 32.44 + 20*log10(fMHz) + 20*log10(dKm);
lossAtPoint = fspl(freqMHz,distanceAtPointKm);
rxAtPoint = txPowerdBm + gtDbi + grDbi - lossAtPoint;
linkTable = table(bandName,freqGHz(:),lossAtPoint(:),rxAtPoint(:), ...
    'VariableNames',{'Band','Frequency_GHz','FSPL_dB','EstimatedRx_dBm'});
disp(linkTable);
figure; tiledlayout(1,2);
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
    plot(distanceKm,fspl(freqMHz(k),distanceKm),'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Path loss (dB)');
legend(bandName,'Location','northwest'); title('Free-Space Path Loss');
nexttile; hold on; grid on;
for k=1:numel(freqMHz)
    rx = txPowerdBm + gtDbi + grDbi - fspl(freqMHz(k),distanceKm);
    plot(distanceKm,rx,'LineWidth',1.5);
end
set(gca,'XScale','log'); xlabel('Distance (km)'); ylabel('Estimated received power (dBm)');
legend(bandName,'Location','southwest'); title('Illustrative Received Power');

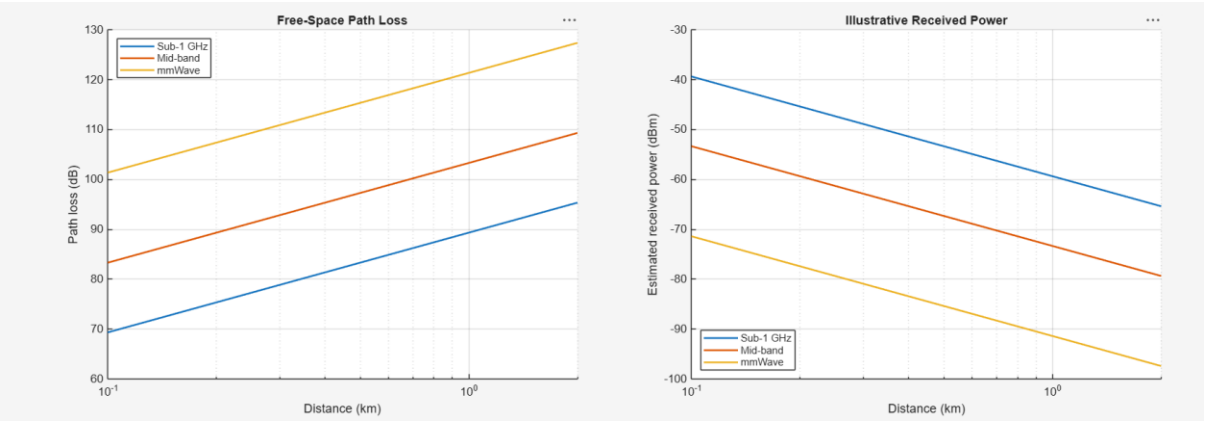
cellRadiusKm = 0.5;
reuseN = [1 3 7 12];
pathLossExponent = 3.5;
reuseDistance = sqrt(3*reuseN) * cellRadiusKm;
sirLinear = (reuseDistance/cellRadiusKm).^pathLossExponent / 6;
sirDb = 10*log10(sirLinear);
reuseTable = table(reuseN(:),reuseDistance(:),sirDb(:), ...
    'VariableNames',{'ReuseClusterN','ApproxInterfererDistance_km','ApproxSIR_dB'});
disp(reuseTable);
figure; plot(reuseN,sirDb,'o-','LineWidth',1.5); grid on;
xlabel('Reuse cluster size N'); ylabel('Approximate SIR (dB)');
title('Simplified Co-Channel SIR versus Reuse Cluster Size');
fprintf('At %.2f km, estimated received powers are:\n',distanceAtPointKm);
disp(table(bandName,rxAtPoint(:),'VariableNames',{'Band','Rx_dBm'}));
```

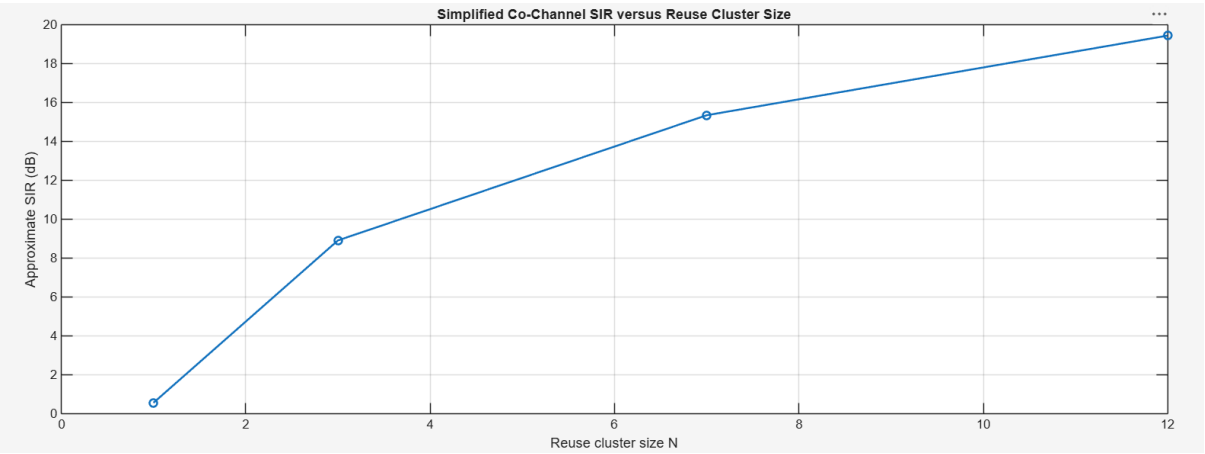
| Band | Frequency_GHz | FSPL_dB | EstimatedRx_dBm |
|---|---|---|---|
| "Sub-1 GHz" | 0.7 | 83.321 | -53.321 |
| "Mid-band" | 3.5 | 97.301 | -67.301 |
| "mmWave" | 28 | 115.36 | -85.363 |

| ReuseClusterN | ApproxInterfererDistance_km | ApproxSIR_dB |
|---|---|---|
| 1 | 0.86603 | 0.56811 |
| 3 | 1.5 | 8.9177 |
| 7 | 2.2913 | 15.357 |
| 12 | 3 | 19.454 |

At 0.50 km, estimated received powers are:

| Band | Rx_dBm |
|---|---|
| "Sub-1 GHz" | -53.321 |
| "Mid-band" | -67.301 |
| "mmWave" | -85.363 |


Free-Space Path Loss
Sub-1 GHz
Mid-band
mmWave
Path loss (dB)
Distance (km)
Illustrative Received Power
Estimated received power (dBm)
Distance (km)


Simplified Co-Channel SIR versus Reuse Cluster Size
Approximate SIR (dB)
Reuse cluster size N